AI ASSISTED CODING

Assignment-14.1

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Lab 14: Web Design Application – AI-Assisted HTML/CSS/JS Generation

Lab Objectives

• Design functional, visually appealing web applications using HTML, CSS, and JavaScript with AI assistance.

• Apply responsive, accessible, and interactive design principles.

• Create practical UI components for real-world web applications.

• Use AI to optimize layout, UX, and performance.

Task 1 – Portfolio Website Design

You are building a personal portfolio website to showcase your work.

Requirements:

• Create sections for About Me, Projects, and Contact.

• Use AI to:

o Suggest color palettes and typography.

o Create a responsive layout with Grid/Flexbox.

o Add smooth scrolling navigation.

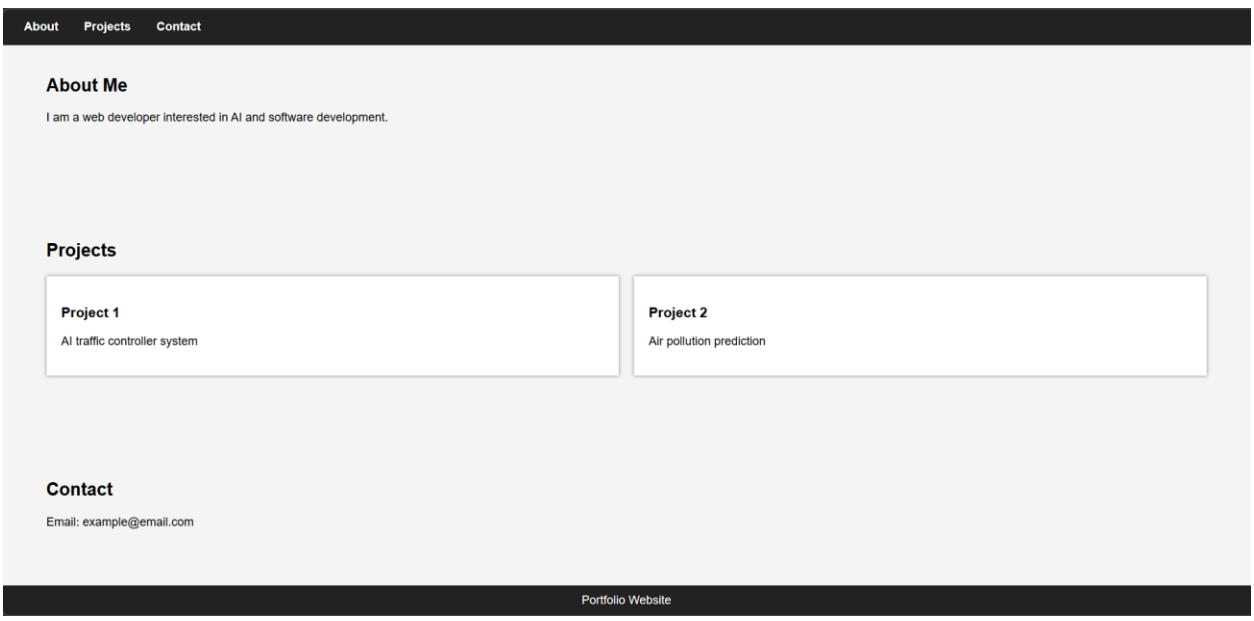
Code:

```html
<!DOCTYPE html>
<html>
<head>
<title>Portfolio</title>

<style>

/* AI suggested color palette */
body{
font-family: Arial, sans-serif;
margin:0;
background: #f4f4f4;
}

/* Navigation bar */
nav{
background: #222;
padding:15px;
position:sticky;
top:0;
}

nav a{
color: white;
margin:15px;
text-decoration:none;
font-weight:bold;
}

/* Smooth scroll */
html{
scroll-behavior:smooth;
}
```

```html
section{
padding:60px;
}

/* Flexbox layout */
.projects{
display:flex;
gap:20px;
flex-wrap:wrap;
}

.project-card{
background: white;
padding:20px;
flex:1;
min-width:200px;
box-shadow:0 0 5px gray;
}

footer{
background: #222;
color: white;
text-align:center;
padding:10px;
}

</style>
</head>

<body>

<nav>
<a href="#about">About</a>
<a href="#projects">Projects</a>
<a href="#contact">Contact</a>
```

```html
</nav>

<section id="about">
<h2>About Me</h2>
<p>I am a web developer interested in AI and software development.</p>
</section>

<section id="projects">
<h2>Projects</h2>

<div class="projects">

<div class="project-card">
<h3>Project 1</h3>
<p>AI traffic controller system</p>
</div>

<div class="project-card">
<h3>Project 2</h3>
<p>Air pollution prediction</p>
</div>

</div>
</section>

<section id="contact">
<h2>Contact</h2>
<p>Email: example@email.com</p>
</section>

<footer>
Portfolio Website
</footer>

</body>
</html>
```

Output:



Task 2 – Online Store Product Page

Design a product display page for an online store.

Requirements:

• Display product image, title, price, and "Add to Cart" button.

• Use AI to:

o Style with BEM methodology.

o Make layout responsive.

o Add hover effects and "Add to Cart" alert.

Code:

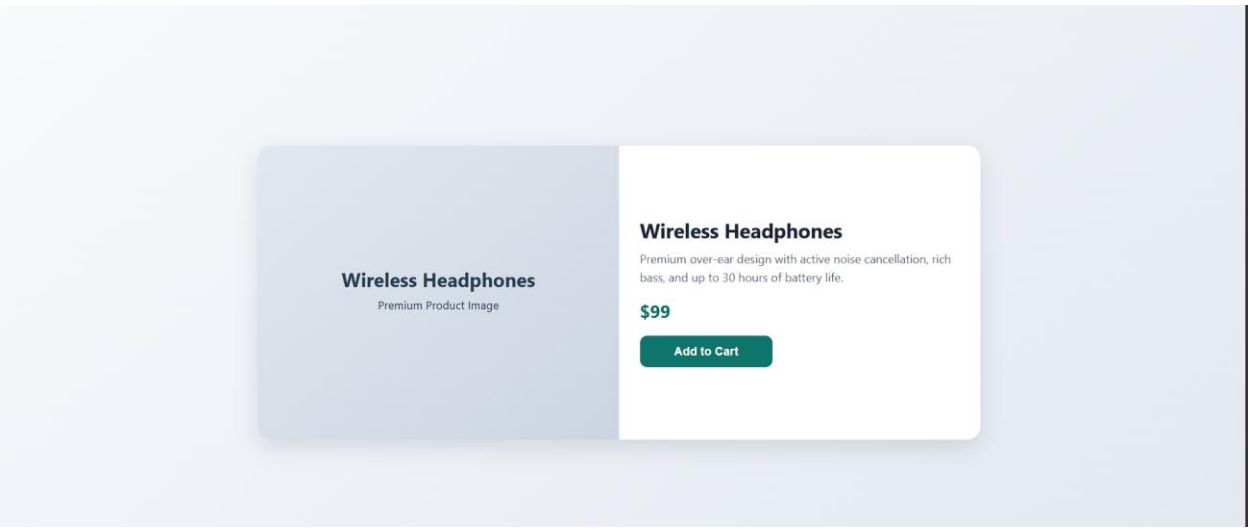
```
.product-card:hover {
    transform: translateY(-6px);
    box-shadow: var(--shadow-hover);
}

.product-card__media {
    overflow: hidden;
    background: #f1f5f9;
}

.product-card__image {
    width: 100%;
    display: block;
    object-fit: cover;
    transition: transform 0.35s ease;
}

.product-card:hover .product-card__image {
    transform: scale(1.04);
}

.product-card__content {
    padding: 22px;
}

.product-card__title {
    margin: 0 0 10px;
    font-size: 1.7rem;
    line-height: 1.25;
}

.product-card__description {
    margin: 0 0 16px;
    color: var(--text-muted);
    line-height: 1.55;
}

.product-card__price {
    margin: 0 0 18px;
    font-size: 1.5rem;
    color: var(--price-color);
    font-weight: 700;
}

.product-card__button {
    width: 100%;
    border: none;
    border-radius: 10px;
```

```html
<!DOCTYPE html>
<html>
<head>
<title>Online Store Product Page</title>
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<style>
:root {
    --bg-start: #f8fafc;
    --bg-end: #e2e8f0;
    --card-bg: #ffffff;
    --text-main: #1f2937;
    --text-muted: #6b7280;
    --price-color: #0f766e;
    --btn-bg: #0f766e;
    --btn-bg-hover: #115e59;
    --shadow-soft: 0 10px 30px rgba(15, 23, 42, 0.12);
    --shadow-hover: 0 18px 35px rgba(15, 23, 42, 0.18);
    --radius-lg: 18px;
}

* {
    box-sizing: border-box;
}

body {
    margin: 0;
    font-family: "Segoe UI", Tahoma, Geneva, Verdana, sans-serif;
    color: var(--text-main);
    background: linear-gradient(140deg, var(--bg-start), var(--bg-end));
}

.store-page {
    min-height: 100vh;
    display: flex;
    align-items: center;
    justify-content: center;
    padding: 28px 16px;
}

.store-page__container {
    width: 100%;
    max-width: 980px;
}

.product-card {
    display: grid;
    grid-template-columns: 1fr;
    background: var(--card-bg);
    border-radius: var(--radius-lg);
    overflow: hidden;
    box-shadow: var(--shadow-soft);
    transition: transform 0.3s ease, box-shadow 0.3s ease;
}
```

```html
    background: var(--btn-bg);
    color: #ffffff;
    font-size: 1rem;
    font-weight: 600;
    padding: 12px 18px;
    cursor: pointer;
    transition: background-color 0.2s ease, transform 0.2s ease;
}

.product-card__button:hover {
    background: var(--btn-bg-hover);
    transform: translateY(-1px);
}

.product-card__button:active {
    transform: translateY(0);
}

@media (min-width: 720px) {
    .product-card {
        grid-template-columns: 1fr 1fr;
    }

    .product-card__content {
        padding: 28px;
        display: flex;
        flex-direction: column;
        justify-content: center;
    }

    .product-card__button {
        width: fit-content;
        min-width: 180px;
    }
}

</style>

<script>

function addCart(){
alert("Product added to cart!");
}

</script>
</head>

<body>
```

```html
<main class="store-page">
    <section class="store-page__container">
        <article class="product-card">
            <div class="product-card__media">
                <img
                    class="product-card__image"
                    src="data:image/svg+xml;utf8,<svg xmlns='http://www.w3.org/2000/svg' width='640' height='520' viewBox='0 0 640 520'><defs><linearGradient id='g' x1='0' y1='0' x2='1' y2='1'><stop offset='0%' stop-color='%23e2e8f0'/><stop offset='10
                    alt="Wireless Headphones"
                >
            </div>

            <div class="product-card__content">
                <h2 class="product-card__title">Wireless Headphones</h2>
                <p class="product-card__description">
                    Premium over-ear design with active noise cancellation, rich bass, and
                    up to 30 hours of battery life.
                </p>
                <p class="product-card__price">$99</p>

                <button class="product-card__button" onclick="addCart()">
                    Add to Cart
                </button>
            </div>
        </article>
    </section>
</main>

</body>
</html>
```

Output:


Wireless Headphones
Premium Product Image
Wireless Headphones
Premium over-ear design with active noise cancellation, rich bass, and up to 30 hours of battery life.
$99
Add to Cart

Task 3 – Event Registration Form

Build an event registration form for a conference.

Requirements:

• Collect name, email, phone number, and session selection.

• Use AI to:

o Add form validation with JavaScript.

o Make the form accessible with labels and ARIA.

o Style with a professional look.

Code:

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Conference Event Registration</title>
  <style>
    :root {
      --page-bg-start: #f1f5f9;
      --page-bg-end: #dbeafe;
      --card-bg: #ffffff;
      --text-main: #1e293b;
      --text-muted: #475569;
      --primary: #0f766e;
      --primary-hover: #115e59;
      --danger: #b91c1c;
      --border: #cbd5e1;
      --focus: #0ea5e9;
      --shadow: 0 16px 36px rgba(15, 23, 42, 0.15);
      --radius: 16px;
    }

    * {
      box-sizing: border-box;
    }

    body {
      margin: 0;
      min-height: 100vh;
      font-family: "Segoe UI", Tahoma, Geneva, Verdana, sans-serif;
      color: var(--text-main);
      background: linear-gradient(145deg, var(--page-bg-start), var(--page-bg-end));
      display: grid;
      place-items: center;
      padding: 20px;
    }

    .registration {
      width: 100%;
      max-width: 680px;
      background: var(--card-bg);
      border-radius: var(--radius);
      box-shadow: var(--shadow);
      overflow: hidden;
    }

    .registration__header {
      padding: 28px 24px 16px;
      background: linear-gradient(135deg, #ecfeff, #f0fdf4);
      border-bottom: 1px solid #e2e8f0;
    }
```

```
.registration__title {
  margin: 0 0 8px;
  font-size: 1.7rem;
  line-height: 1.2;
}

.registration__subtitle {
  margin: 0;
  color: var(--text-muted);
  line-height: 1.5;
}

.registration__form {
  padding: 20px 24px 26px;
  display: grid;
  gap: 16px;
}

.registration__row {
  display: grid;
  grid-template-columns: 1fr;
  gap: 16px;
}

.field {
  display: grid;
  gap: 8px;
}

.field__label {
  font-weight: 600;
  color: #0f172a;
}

.field__required {
  color: var(--danger);
}

.field__input,
.field__select {
  width: 100%;
  padding: 11px 12px;
  border: 1px solid var(--border);
  border-radius: 10px;
  font-size: 0.98rem;
  color: var(--text-main);
  background: #ffffff;
```

```
  transition: border-color 0.2s ease, box-shadow 0.2s ease;
}

.field__input:focus,
.field__select:focus {
  outline: none;
  border-color: var(--focus);
  box-shadow: 0 0 0 3px rgba(14, 165, 233, 0.2);
}

.field__error {
  min-height: 18px;
  margin: 0;
  color: var(--danger);
  font-size: 0.86rem;
}

.field--invalid .field__input,
.field--invalid .field__select {
  border-color: var(--danger);
  box-shadow: 0 0 0 3px rgba(185, 28, 28, 0.14);
}

.registration__actions {
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 12px;
  margin-top: 4px;
  flex-wrap: wrap;
}

.registration__button {
  border: none;
  border-radius: 10px;
  background: var(--primary);
  color: #ffffff;
  padding: 12px 18px;
  font-size: 1rem;
  font-weight: 600;
  cursor: pointer;
  transition: transform 0.2s ease, background-color 0.2s ease;
}
```

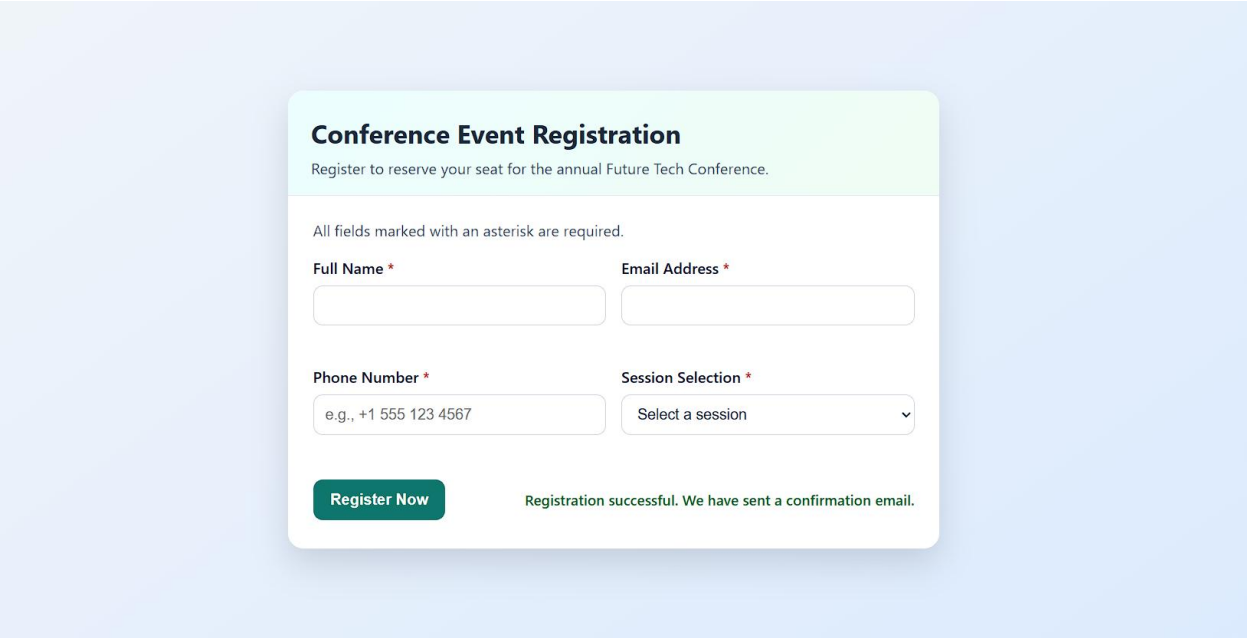
```
    .registration__button:hover {
      background: var(--primary-hover);
      transform: translateY(-1px);
    }

    .registration__button:active {
      transform: translateY(0);
    }

    .registration__status {
      margin: 0;
      color: #166534;
      font-size: 0.95rem;
      font-weight: 600;
      min-height: 18px;
    }

    @media (min-width: 700px) {
      .registration__row {
        grid-template-columns: 1fr 1fr;
      }

      .registration__form {
        padding: 24px 26px 30px;
      }
    }
  </style>
</head>
<body>
  <section class="registration" aria-labelledby="registration-title">
    <header class="registration__header">
      <h1 class="registration__title" id="registration-title">Conference Event Registration</h1>
      <p class="registration__subtitle">Register to reserve your seat for the annual Future Tech Conference.</p>
    </header>

    <form class="registration__form" id="registrationForm" novalidate aria-describedby="formInstructions">
      <p id="formInstructions" class="registration__subtitle">All fields marked with an asterisk are required.</p>

      <div class="registration__row">
        <div class="field" id="nameField">
          <label class="field__label" for="fullName">Full Name <span class="field__required" aria-hidden="true">*</span></label>
          <input class="field__input" type="text" id="fullName" name="fullName" autocomplete="name" required aria-required="true" aria-describedby="nameError">
          <p class="field__error" id="nameError" role="alert" aria-live="polite"></p>
        </div>
```

```
        <div class="field" id="emailField">
          <label class="field__label" for="email">Email Address <span class="field__required" aria-hidden="true">*</span></label>
          <input class="field__input" type="email" id="email" name="email" autocomplete="email" required aria-required="true" aria-describedby="emailError">
          <p class="field__error" id="emailError" role="alert" aria-live="polite"></p>
        </div>
      </div>

      <div class="registration__row">
        <div class="field" id="phoneField">
          <label class="field__label" for="phone">Phone Number <span class="field__required" aria-hidden="true">*</span></label>
          <input class="field__input" type="tel" id="phone" name="phone" autocomplete="tel" required aria-required="true" aria-describedby="phoneError" placeholder="e.g., +1 555 123 4567">
          <p class="field__error" id="phoneError" role="alert" aria-live="polite"></p>
        </div>

        <div class="field" id="sessionField">
          <label class="field__label" for="session">Session Selection <span class="field__required" aria-hidden="true">*</span></label>
          <select class="field__select" id="session" name="session" required aria-required="true" aria-describedby="sessionError">
            <option value="">Select a session</option>
            <option value="ai-in-business">AI in Business</option>
            <option value="cloud-architecture">Cloud Architecture</option>
            <option value="cybersecurity-trends">Cybersecurity Trends</option>
            <option value="product-leadership">Product Leadership</option>
          </select>
          <p class="field__error" id="sessionError" role="alert" aria-live="polite"></p>
        </div>
      </div>

      <div class="registration__actions">
        <button class="registration__button" type="submit">Register Now</button>
        <p class="registration__status" id="formStatus" aria-live="polite"></p>
      </div>
    </form>
  </section>

  <script>
    const form = document.getElementById("registrationForm");
    const status = document.getElementById("formStatus");

    const fields = {
      fullName: {
        input: document.getElementById("fullName"),
        wrapper: document.getElementById("nameField"),
        error: document.getElementById("nameError"),
        validate: function (value) {
          if (!value.trim()) return "Name is required.";
```

```
      if (!/^[A-Za-z ]{2,}$/.test(value.trim())) return "Enter at least 2 letters for your name.";
      return "";
    }
  },
  email: {
    input: document.getElementById("email"),
    wrapper: document.getElementById("emailField"),
    error: document.getElementById("emailError"),
    validate: function (value) {
      if (!value.trim()) return "Email is required.";
      if (!/^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(value.trim())) return "Enter a valid email address.";
      return "";
    }
  },
  phone: {
    input: document.getElementById("phone"),
    wrapper: document.getElementById("phoneField"),
    error: document.getElementById("phoneError"),
    validate: function (value) {
      const clean = value.replace(/[^\d+]/g, "");
      if (!value.trim()) return "Phone number is required.";
      if (!/^\+?\d{10,15}$/.test(clean)) return "Enter a valid phone number (10-15 digits).";
      return "";
    }
  },
  session: {
    input: document.getElementById("session"),
    wrapper: document.getElementById("sessionField"),
    error: document.getElementById("sessionError"),
    validate: function (value) {
      if (!value) return "Please select a session.";
      return "";
    }
  }
};

function setFieldState(field, message) {
  field.error.textContent = message;
  const hasError = Boolean(message);
  field.wrapper.classList.toggle("field--invalid", hasError);
  field.input.setAttribute("aria-invalid", hasError ? "true" : "false");
}
```

```
function validateField(key) {
  const field = fields[key];
  const message = field.validate(field.input.value);
  setFieldState(field, message);
  return !message;
}

Object.keys(fields).forEach(function (key) {
  fields[key].input.addEventListener("input", function () {
    validateField(key);
  });

  fields[key].input.addEventListener("blur", function () {
    validateField(key);
  });
});

form.addEventListener("submit", function (event) {
  event.preventDefault();
  status.textContent = "";

  const keys = Object.keys(fields);
  const firstInvalidKey = keys.find(function (key) {
    return !validateField(key);
  });

  if (firstInvalidKey) {
    fields[firstInvalidKey].input.focus();
    status.textContent = "Please fix the highlighted fields.";
    return;
  }

  status.textContent = "Registration successful. We have sent a confirmation email.";
  form.reset();
  keys.forEach(function (key) {
    setFieldState(fields[key], "");
  });
});
</script>
</body>
</html>
```

Output:


Conference Event Registration
Register to reserve your seat for the annual Future Tech Conference.
All fields marked with an asterisk are required.
Full Name *
Email Address *
Phone Number *
e.g., +1 555 123 4567
Session Selection *
Select a session
Register Now
Registration successful. We have sent a confirmation email.

Task 4 – Resume Website

Build a one-page online resume.

Requirements:

• Sections: Profile Summary, Skills, Education, Experience.

• Use AI to:

o Propose professional typography styles.

o Align sections using CSS Grid.

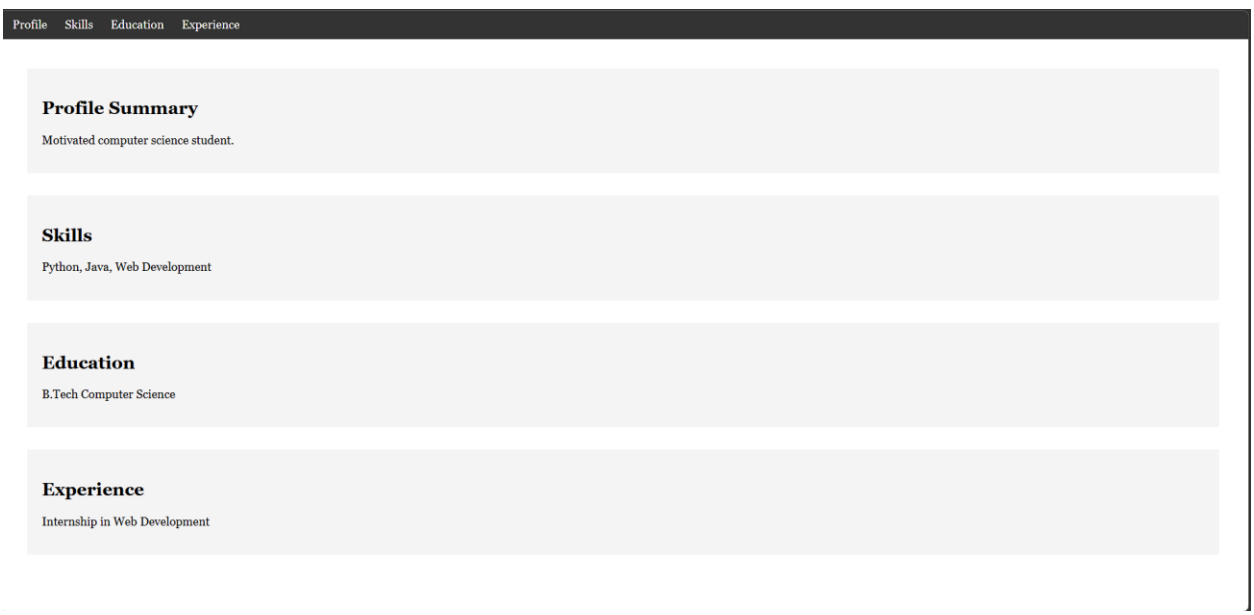
o Add smooth scroll navigation between sections.

Code:

```
<!DOCTYPE html>
<html>
<head>

<title>Online Resume</title>

<style>

body{
font-family:Georgia;
margin:0;
}

/* Navigation */

nav{
background:#333;
padding:10px;
}

nav a{
color:white;
margin:10px;
text-decoration:none;
}

/* Smooth scroll */

html{
scroll-behavior:smooth;
}

/* Grid Layout */

.container{
display:grid;
grid-template-columns:1fr;
gap:30px;
padding:40px;
}

section{
background:#f4f4f4;
padding:20px;
}

</style>
</head>
```

```html
<body>

<nav>
<a href="#profile">Profile</a>
<a href="#skills">Skills</a>
<a href="#education">Education</a>
<a href="#experience">Experience</a>
</nav>

<div class="container">

<section id="profile">
<h2>Profile Summary</h2>
<p>Motivated computer science student.</p>
</section>

<section id="skills">
<h2>Skills</h2>
<p>Python, Java, Web Development</p>
</section>

<section id="education">
<h2>Education</h2>
<p>B.Tech Computer Science</p>
</section>

<section id="experience">
<h2>Experience</h2>
<p>Internship in Web Development</p>
</section>

</div>

</body>
</html>
```

Output:



Task 5 – Restaurant Menu Page

Design a digital menu page for a restaurant.

Requirements:

• Display menu items with name, description, and price.

• Use AI to:

o Suggest a food-friendly color palette.

o Design a responsive card-based layout.

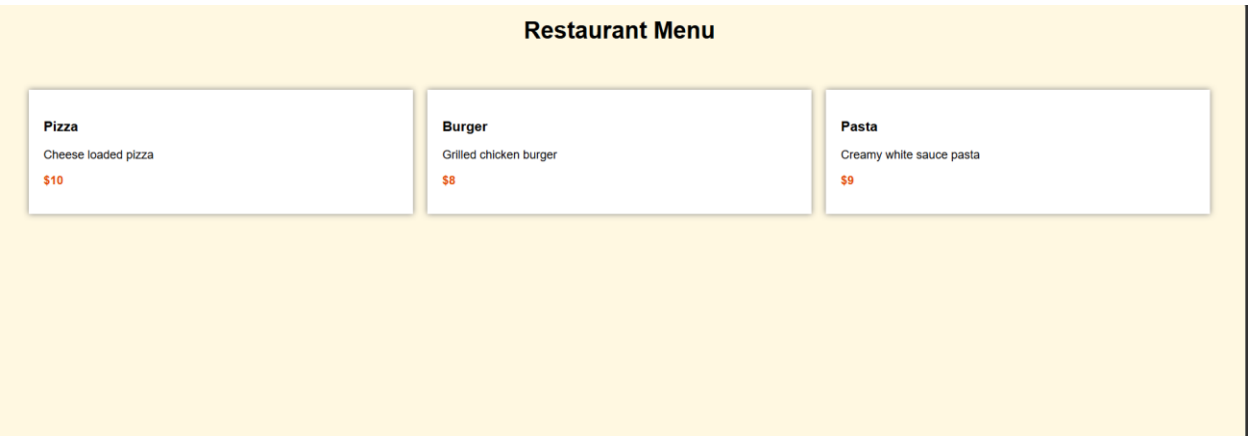
o Add hover effects on menu items.

Code:

```html
<!DOCTYPE html>
<html>
<head>
<title>Restaurant Menu</title>

<style>

body{
font-family:Arial;
background:#fff8e1;
}

/* Grid layout */

.menu{
display:grid;
grid-template-columns:repeat(auto-fit,minmax(200px,1fr));
gap:20px;
padding:40px;
}

.card{
background:white;
padding:20px;
box-shadow:0 0 8px gray;
transition:transform 0.3s;
}

/* Hover effect */

.card:hover{
transform:scale(1.05);
}

.price{
color:#e65100;
font-weight:bold;
}

</style>
</head>

<body>

<h1 style="text-align:center">Restaurant Menu</h1>

<div class="menu">
```

```html
<div class="card">
<h3>Pizza</h3>
<p>Cheese loaded pizza</p>
<p class="price">$10</p>
</div>

<div class="card">
<h3>Burger</h3>
<p>Grilled chicken burger</p>
<p class="price">$8</p>
</div>

<div class="card">
<h3>Pasta</h3>
<p>Creamy white sauce pasta</p>
<p class="price">$9</p>
</div>

</div>

</body>
</html>
```

Output:


Restaurant Menu
Pizza
Cheese loaded pizza
$10
Burger
Grilled chicken burger
$8
Pasta
Creamy white sauce pasta
$9